# SIMATS ENGINEERING
## ASSESSMENT TOOL 1

**COURSE NAME: Artificial Intelligence** **COURSE CODE: CSA1708**

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**COURSE OUTCOME COVERED**

***CO1:*** Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

*Assessment Tool Weightage: 50%*

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**QUESTIONS: 5**

**Duration : 1 Hour**
**Total Marks:50**

Annexure A

**Analytical Problem Solving**

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***Code of Conduct:***

*I **Lokesh S (192511037)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

Annexure A

# Analytical Problem Solving

**1.** Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

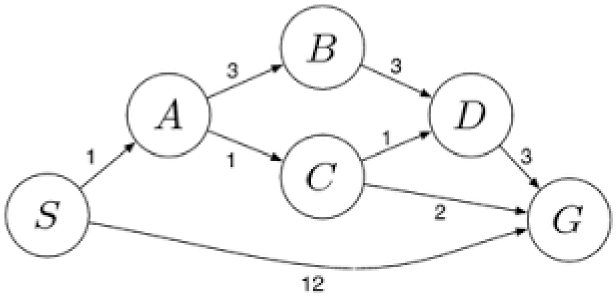
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
   i. What are the percept's for this agent?
   ii. Characterize the operating environment.
   iii. What are the actions the agent can take?
   iv. How can one evaluate the performance of the agent?
   v. What sort of agent architecture do you think is most suitable for this agent? Why?

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

**4.** A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

   The delivery network is represented as a weighted graph:




**RUBRICS FOR CALCULATION:**

| Criteria | Weightage | Level 4 (Exemplary) | Level 3 (Proficient) | Level 2 (Developing) | Level 1 (Beginning) |
|---|---|---|---|---|---|
| Problem Understanding | 20% | Clearly interprets problem, identifies all parameters and requirements accurately | Understands problem with minor gaps | Partial understanding; misses key elements | Misinterprets or unable to understand problem |
| Method Selection | 20% | Selects most appropriate method/formula with strong justification | Selects correct method with minor errors | Inappropriate or partially correct method | Unable to select method |
| Solution Process | 25% | Logical, systematic steps with correct approach throughout | Mostly correct steps with minor mistakes | Disorganized steps; multiple errors | No clear process |
| Accuracy of Calculation | 20% | All calculations accurate; correct final answer | Minor calculation errors | Frequent errors; incorrect final answer | No valid calculations |
| Interpretation of Results | 15% | Clearly interprets results with units and engineering relevance | Basic interpretation with minor omissions | Limited or unclear interpretation | No interpretation |
| Problem Understanding | 20% | Clearly interprets problem, identifies all parameters and requirements accurately | Understands problem with minor gaps | Partial understanding; misses key elements | Misinterprets or unable to understand problem |

**Question 1: Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.**

**Problem Statement**
You are given two jugs:

- Jug A = 4 gallons
- Jug B = 3 gallons

The objective is to obtain **exactly 2 gallons** in the 4-gallon jug using only:

- Filling a jug completely
- Emptying a jug
- Pouring water from one jug to another

**State Representation**
Represent each state as:
**(A, B)**
Where:

- A = Amount of water in the 4-gallon jug
- B = Amount of water in the 3-gallon jug

Initial State:
**(0,0)**
Goal State:
**(2,x)** where x can be any value.

**Solution Steps**

| **Step** | **Action** | **State (4-gallon, 3-gallon)** |
|---|---|---|
| Initial | Both empty | (0,0) |
| 1 | Fill 3-gallon jug | (0,3) |
| 2 | Pour into 4-gallon jug | (3,0) |
| 3 | Fill 3-gallon jug again | (3,3) |
| 4 | Pour into 4-gallon until full | (4,2) |
| 5 | Empty 4-gallon jug | (0,2) |
| 6 | Pour remaining 2 gallons into 4-gallon jug | **(2,0)** |

**Goal Achieved**
The 4-gallon jug contains exactly **2 gallons**.

**State Space Diagram**
(0,0)
  |
Fill 3
  |
(0,3)
  |
Pour

|
(3,0)
|
Fill 3
|
(3,3)
|
Pour
|
(4,2)
|
Empty 4
|
(0,2)
|
Pour
|
(2,0) ✓

**Question 2: Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.**

- **vi. What are the percept's for this agent?**
- **vii. Characterize the operating environment.**
- **viii. What are the actions the agent can take?**
- **ix. How can one evaluate the performance of the agent?**
- **x. What sort of agent architecture do you think is most suitable for this agent? Why?**

The Mars Rover is an autonomous intelligent agent that explores Mars, collects data, and performs scientific experiments without continuous human control.

**i) Percepts**

Percepts are the information collected from sensors.
The Mars Rover perceives:

- Surface images
- Rocks and soil composition
- Temperature
- Atmospheric pressure
- Wind speed
- Terrain obstacles
- GPS/location coordinates
- Battery level
- Wheel condition
- Radiation level

**ii) Operating Environment**

| **Property** | **Description** |
|---|---|
| Observable | Partially observable |
| Deterministic | No (uncertain terrain) |

| Property | Description |
|---|---|
| Dynamic | Yes |
| Sequential | Yes |
| Continuous | Yes |
| Multi-agent | No |

**iii) Actions**
The Mars Rover can:
- Move forward
- Turn left/right
- Stop
- Capture images
- Drill rocks
- Collect soil samples
- Analyze samples
- Send data to Earth
- Recharge using solar panels
- Avoid obstacles

**iv) Performance Measures**
Performance is evaluated using:
- Number of successful samples collected
- Distance covered
- Scientific discoveries
- Battery efficiency
- Communication success
- Mission completion
- Safe navigation
- Obstacle avoidance accuracy

**v) Suitable Agent Architecture**
**Model-Based Goal-Oriented Agent**
**Reason:**
The Mars Rover operates in a partially observable and dynamic environment. It maintains an internal model of the environment, plans actions to achieve mission goals, and adapts based on sensor data.

**Question 3: Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column. The objective is to place 8 queens on an 8 × 8 chessboard so that no queen attacks another.**

**Problem Formulation**
**Initial State**
Empty chessboard.

**State Space**
A state is a board configuration with queens placed in valid positions.

**Successor Function**
Place one queen in the next column without violating constraints.

**Goal Test**
The goal is achieved when:

- Eight queens are placed.
- No two queens share:
  - Same row
  - Same column
  - Same diagonal

**Path Cost**
Each queen placement costs 1.
Total cost:
**8**

**Search Space**
Empty Board

↓

Place Queen in Column 1

↓

Place Queen in Column 2

↓

...

↓

Place Queen in Column 8

↓

Goal State

**State Representation**
Each queen position can be represented as:
(Row1, Row2, Row3, Row4, Row5, Row6, Row7, Row8)
Example Solution:
(1,5,8,6,3,7,2,4)

**Suitable Search Strategy**
Backtracking Search (Depth-First Search with Constraint Checking).

**Advantages**

- Reduces unnecessary exploration.
- Efficient memory usage.
- Guarantees a valid solution.

**Question 4: A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.**

A customer uses the OLA app to choose different cab types (Mini, Micro, Sedan, Prime, Shared, etc.) to travel from one location to another.

**Type of Problem-Solving Agent**
**Goal-Based Agent**

**Reason**
The customer's goal is to reach the destination. The agent evaluates available cab options based on user preferences (cost, comfort, travel time, etc.) and selects the most suitable one.

**Pseudocode**

```
START

Input source location
Input destination
Input customer preference

Search available cabs

IF no cab available
   Display "No Cab Available"
ELSE
   Display available cab types
   Select best cab based on preference
   Calculate fare
   Confirm booking
   Assign driver
   Start trip
ENDIF

Reach destination

END
```

**Working Process**
Customer

↓

Enter Source

↓

Enter Destination

↓

Search Available Cabs

↓

Compare Fare & Comfort

↓

Select Best Cab

↓

Booking Confirmation

↓

Driver Assigned

↓

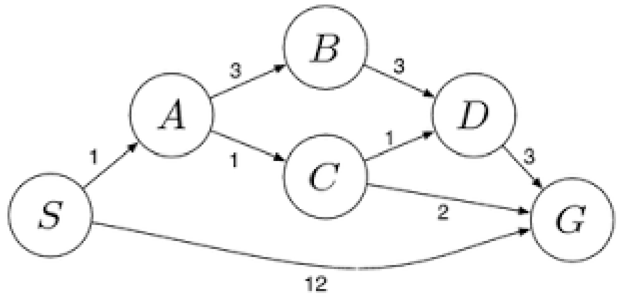
Trip Starts

↓

Destination Reached

**Question 5: A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.**

**The delivery network is represented as a weighted graph:**




The delivery network is represented by the weighted graph shown in the question. The edge costs are:

- S → A = 1
- S → G = 12
- A → B = 3
- A → C = 1
- B → D = 3
- C → D = 1
- C → G = 2
- D → G = 3

The objective is to find the least-cost path from **S** to **G** using Uniform Cost Search.

**Step-by-Step UCS Execution**
**Step 1**
Start at:
S (Cost = 0)
Expand:
A (1)
G (12)

**Step 2**
Choose lowest cost node.
A (1)
Expand:
B (4)
C (2)

**Step 3**
Choose
C (2)
Expand:
D (3)
G (4)

**Step 4**
Choose
D (3)
Expand:
G (6)

Existing G already has cost 4.
Keep lower cost.

**Step 5**
Choose
G (4)
Goal reached.

**Search Table**

| **Expanded Node** | **Path Cost** |
|---|---|
| S | 0 |
| A | 1 |
| C | 2 |
| D | 3 |
| G | 4 |

**Least-Cost Path**
S
↓

A

↓

C

↓

G

**Total Cost**
S → A = 1

A → C = 1

C → G = 2

Total Cost = 4